

GarageBand VR Drums

Testing plan for interactive prototype 1

GarageBand VR Drums is an interactive virtual drumming experience that lets users play a drum kit, record a short performance, and replay it. Instead of using a conventional music app, it uses XR to place the user in a virtual drum space where playing is performed through spatial interaction with drumsticks. The prototype aims to make basic drumming feel engaging and understandable through a simple, game-like flow.

Testing Objective

From the concept above, I have identified an assumption that first-time users can understand the core flow of the experience after receiving the required simulator controls: entering the drum stage, selecting a drumstick, playing the drums, and using the recording controls. I also need to validate whether the placement and feedback of the drums and recording control panel make these actions clear and understandable. This test aims to discover whether users can complete the main interaction flow with little or no assistance, where they hesitate or become confused, and whether the drum layout and recording controls are clear enough for a first-time user. It also aims to identify any unexpected actions, misunderstandings, or usability issues that may not be apparent when testing the prototype alone, as participants who were not involved in creating the prototype may approach and interpret the interaction differently. Success will be measured through task completion, amount of assistance required, observed errors or hesitation, unexpected interactions, and participant feedback.

Testing Methodologies

This testing plan uses moderated task-based usability testing with direct observation and short post-test questions to evaluate a Unity prototype. Each participant will complete the same core tasks while the evaluator records task success, assistance needed, observed behaviour, and relevant participant comments. Technical simulator controls will be explained through an instruction panel before the test. The recording interface itself will not be explained so its clarity can be evaluated.

Prototype description/requirements

The prototype was designed to provide the complete high-level experience needed to test the main drumming and recording flow. It includes a virtual drum stage and drum seat, movement controls, selectable left and right drumsticks, playable drums with audio feedback, plus a control panel for recording, stopping, and replaying a performance. A short "How to Play" panel explains required Unity/XR simulator controls: W/A/S/D to move, clicking the drum stage to sit, T/Y to select a drumstick, and cursor movement to move the selected drumstick.

Data collection method

During the testing process, the evaluator will audio-record each session, with the participant's consent, so that participant comments and reactions are not missed while the evaluator observes their interaction with the prototype. The evaluator will also complete a structured observation sheet for each task, recording the participant number (e.g., Participant 1 or Participant 2), what the participant did, the result, the level of assistance needed, any direct participant quotes, and additional observer comments. After testing, the audio recordings will be transcribed and analysed alongside the observation notes to identify key findings that can inform improvements to the prototype. Participants will be labelled P1, P2, and so on rather than by name.

Testing Setup

Before class, the Unity build will be opened and checked for working movement, drumstick selection, drum audio, the drum seat, and the Record/Stop/Play controls. The evaluator will ensure there is enough space for a participant to use the computer comfortably. At least five classmates or tutors will be invited to participate. Each testing session will take approximately 10 to 11 minutes.

Testing process: (also considering the schedule/time)

1. Welcome and consent (approximately 1 minute): "Thank you for testing my GarageBand VR Drums prototype. So in this testing process I will give you a few tasks to complete using the prototype. Please try to complete each task on your own first. If you have any questions, feel free to ask, but try to figure it out yourself before asking for help. Please think aloud where possible as well. Is it okay if I record this interview?"
2. Explain simulator controls (approximately 1 minute): The evaluator will ask the participant to read the How to Play panel and clarify only the technical controls needed to operate the Unity/XR simulator. The evaluator will not explain the functions of the Record, Stop, or Play buttons, as these will be evaluated during the tasks.
3. Task completion and observation (approximately 5.5 minutes):
 - **Task 1 – Enter the drum seat (1 minute):** "Please move to the drum stage and sit at the drum kit." Observe whether the participant can identify the drum stage and enter the drum seat without assistance.
 - **Task 2 – Start recording (30 seconds):** "Please start recording your performance." Observe whether the participant can find and understand the Record button without instruction.
 - **Task 3 – Play the drums (3 minutes):** "Please select and use each drumstick, one at a time, to play a short rhythm using the drums." Observe whether the participant can select the drumsticks, move them towards the drums, and receive clear audio feedback.

- **Task 4 – Stop recording (30 seconds):** “Please stop the recording.” Observe whether the participant can find and understand the Stop Recording control without instruction.
 - **Task 5 – Play recording (30 seconds):** “Please play back your recording.” Observe whether the participant can identify and use the Play Recording control to replay their performance without further instruction.
4. Short interview and close (Approximately 3 minutes): Ask: “Walk me through your thoughts, what are you thinking?”, “What was the easiest task?”, “What was the most difficult or confusing task?”, and “What would you change to improve your experience?”

	0: Fail	1: Pass	2: Good	3: Excellent
Prototype Development and Alignment	The Unity prototype is non-functional or missing. It does not align with the original concept or the testing plan.	The prototype is minimally functional but may be unstable, contain significant bugs or does not function in an expected manner. It aligns weakly with the original concept and only addresses a few basic testing goals. It is not complete for testing purpose.	The prototype is functional, complete and stable, allowing for testing of key user tasks. It aligns well with the original concept and is horizontally complete enough to facilitate the testing plan.	The prototype is highly functional, robust, and polished. It provides a comprehensive impression of the concept and appears complete, with well-implemented interactions.
Testing Plan & Methodology	No testing plan is provided, or it is vague and impractical. There is no clear goal for the testing session.	A testing plan is provided, but it lacks detail or clarity. It is difficult for another person to follow, and the questions or tasks may not effectively validate the prototype.	A clear and well-structured testing plan is provided. It outlines specific, testable interactions and identifies the assumptions to be validated. The plan is robust enough for another person to follow and collect meaningful results.	A comprehensive and insightful testing plan is provided. It not only clearly outlines testable interactions and assumptions but also includes a well-justified methodology (e.g., specific metrics, data collection forms). The plan is exceptionally clear and designed to generate rich, actionable data.
Execution of Testing & Data Collection	The student does not participate in the in-class testing session or fails to log any results. No data has been collected.	The student participates in testing, but the data collected is minimal, superficial or poorly organised. The logged results are incomplete or disorganised.	The student actively participates in the in-class testing, engaging with peers and the tutor. They successfully collect and log testing results in a clear and organised manner	The student is highly engaged and proactive during the testing session, and collecting a decent volume of high-quality, actionable data. The results are meticulously logged, well-organised, and show a clear understanding of the data's purpose.